

# TEJAS TIKKAS

7821964097 • tejastikkas2545@gmail.com • [Github](#) • [LinkedIn](#) • [Portfolio](#)

## SUMMARY

High-performing Computer Science Engineer (**9.13 CGPA**) specializing in **Generative AI, Multi-Agent Systems, and Production RAG**. Proven ability architecting agentic development platforms with local LLMs (Ollama), production SaaS tools using Gemini and Google Cloud APIs, and full-stack web/mobile ecosystems. Winner of 6 technical competitions and Founder of PyVerse Coding Club.

## TECHNICAL SKILLS

|                                      |                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>AI &amp; Agentic Systems:</b>     | Multi-Agent Systems, Production RAG, Ollama (Local LLMs), Gemini API, LangChain, ChromaDB, Prompt Engineering. |
| <b>Machine &amp; Deep Learning:</b>  | PyTorch, TensorFlow, OpenCV, Scikit-learn, CNNs, Transformers, DETR, ViTs, U-Net, Transfer Learning.           |
| <b>Full-Stack &amp; Backend:</b>     | Python (FastAPI, Flask), React 19, TypeScript, REST APIs, OAuth 2.0, Vite, Tailwind CSS, SQL.                  |
| <b>Tools &amp; Cloud Deployment:</b> | Git, GitHub Actions, Docker, Edge Impulse, Power BI (DAX), Linux, Vercel, Supabase.                            |

## KEY AI & ENGINEERING PROJECTS

### DevForge AI — Multi-Agent Software Engineering Platform with Production RAG 2026

Ollama, Multi-Agent Orchestration, Vector Embeddings, ChromaDB, FastAPI, Python

- Architected hierarchical **multi-agent workflow** assigning distinct phases (Planning, Code Gen, Review, Debugging, Automated Testing) across local LLMs (*Qwen2.5-Coder, Devstral, DeepSeek-Coder-V2*).
- Implemented an in-memory **Production RAG pipeline** indexing multi-file codebases with semantic vector search, reducing context hallucination by **>45%**.
- Engineered an automated benchmarking suite to evaluate code quality, execution latency, and GPU/RAM consumption across single vs. multi-model setups.

### Prompt2Form — Conversational AI Google Form Generator ([prompt2form.tejastikkas.site](#)) 2026

Google Gemini API, Google Forms REST API v1, FastAPI, React 19, TypeScript, OAuth 2.0, Vercel

- Engineered an automated SaaS application converting unstructured natural language prompts into fully populated, schema-validated Google Forms in under **5 seconds**.
- Implemented **conversational in-place live editing** via Google Forms batchUpdate API, allowing iterative schema mutations without duplicating files.
- Integrated Google Workspace OAuth 2.0 authentication and custom Pydantic schemas for reliable JSON generation.

### PyVerse Learning Ecosystem — Custom Mobile App & Exam Portal ([mitpyverse.tejastikkas.site](#)) 2025 – 2026

Flutter / React, Python, REST APIs, PostgreSQL, Role-Based Access Control

- Designed and deployed an integrated mobile application and online examination portal serving **100+ active students** of the MIT PyVerse Coding Club.
- Built automated coding assessment evaluation, anti-cheat test environments, student score tracking, and real-time rank dashboards.

### AI-Driven ESR Measurement & Diagnostic Healthcare Monitoring (Ongoing Research) Research Project

Computer Vision, OpenCV, IoT, ESP32, Time-Series Image Processing

- Developing automated Erythrocyte Sedimentation Rate estimation system combining Computer Vision segmentation with IoT sensors to improve medical diagnostic speed and consistency.

## INTERNSHIP EXPERIENCE

### AI/ML Intern & Team Lead — ByteUprise 6 Months

- Led a 5-member engineering team designing and deploying production machine learning models for real-world enterprise use cases.
- Improved model throughput by **35%** via pipeline pruning and feature engineering; maintained CI/CD workflows and code quality standards.

### Python & Data Analytics Intern — CodeElevate Technologies 6 Weeks

- Constructed automated data cleaning and EDA scripts using NumPy and Pandas; delivered interactive Power BI reporting dashboards using DAX formulas.

## LEADERSHIP & HONORS

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|----------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Founder &amp; Lead:</b> MIT PyVerse Club (Mentored 100+ students) | <b>2nd Prize (₹15,000 Cash):</b> Future Founders, Shivaji College |
| <b>1st Prize:</b> Clone Logic Hackathon, SVERI Pandharpur            | <b>2nd Prize:</b> CodeZen (Invento 2K25, DY Patil Kolhapur)       |
| <b>1st Rank:</b> Mini Hackathon, MITCORER Barshi                     | <b>2nd Prize:</b> Project Expo, BIGCE Solapur                     |

## EDUCATION

### B.Tech in Computer Science & Engineering — MITCORER, Barshi

Diploma in Computer Engineering — Govt. Polytechnic, Arvi

**CGPA: 9.13** (Sem 1–4) | SGPA: 9.04 (Sem 5)

**80.86%** (Distinction) | 2024